Behavioral Analytics: What does that even mean?

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Abstract. Behavior is a foundational yet often implicitly defined concept in process mining. Although techniques such as process discovery, conformance checking, and performance analysis are fundamentally grounded in behavioral aspects of event data, the notion of "behavior" is frequently used without explicit conceptualization. This lack of clarity can hinder methodological consistency, complicate comparisons across studies, and obscure the assumptions underlying analytical approaches. This paper investigates how behavior and behavioral analytics are understood in process mining literature, focusing on contributions from the Event Data and Behavioral Analytics workshop at the International Conference on Process Mining. Through a qualitative analysis of workshop publications, we uncover implicit definitions and interpretations of behavior and behavioral analytics, identifying nine major themes that play an important role. Rather than prescribing a universal definition, this work provides an intermediary step toward clarifying the conceptual boundaries of behavior and behavioral analytics, with the aim of fostering discussion, making assumptions explicit, and identifying directions for future theory-building. Ultimately, this exploratory mapping supports more coherent methodologies and lays the groundwork for cumulative knowledge building in the field.

Keywords: Behavior · Behavioral Analytics · Process Mining.

1 Introduction

Process mining has established itself as an important discipline for analyzing and improving business processes. By leveraging the digital footprints left in information systems, process mining enables the discovery of how the process was actually executed, the detection of deviations from prescribed models, and the generation of insights to support data-driven decision-making [1]. At the core of all these analyses lies the concept of behavior, a term that is frequently used but rarely conceptualized within the process mining literature.

In a process mining context, behavior typically refers to the observable dynamics of a process as it unfolds over time: sequences of activities, branching

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choices, repetitions, and timing patterns [1]. Behavioral aspects are central to many process mining techniques, including process discovery [3,12], conformance checking [13,16], and performance analysis [3]. Yet, despite its foundational role, the notion of behavior is often assumed rather than explicitly defined. This lack of conceptual clarity can hinder the development of more precise analysis techniques, for example, when distinguishing between individual behavior (user interactions) and collective behavior (team dynamics) in process execution. It can also complicate communication across research efforts, as different studies may implicitly use divergent interpretations of behavior, making it difficult to compare findings or build on each other's work.

Recognizing the importance of behavioral aspects in process mining, the workshop on Event Data and Behavioral Analytics (EdbA) at the International Conference on Process Mining (ICPM) has emerged as a dedicated venue for exploring behavior-centric perspectives[5]. This workshop explicitly focuses on understanding, modeling, and analyzing behavior within event data, reflecting a growing awareness that behavior is more than just control-flow. However, even within the EdbA community, the term *behavior* or *behavioral analytics* is used in diverse ways, often without a shared conceptual grounding.

This paper takes an intermediary step toward clarifying how behavior and behavioral analytics are understood in process mining, with a focus on contributions from the EdbA workshop. As the only venue dedicated to behavioral perspectives in process mining, EdbA offers a concentrated corpus for our research question. Rather than prescribing a universal definition, we aim to uncover implicit meanings, clarify conceptual boundaries, and identify opportunities to strengthen the theoretical foundations of behavior in process mining. Our goal is to inform and initiate broader discussion, recognizing that any consolidated theory will require multiple perspectives over time.

To this end, we conducted a qualitative analysis of all EdbA papers, using open coding to identify recurring themes, interpretations, and assumptions. While this provides a structured view of current conceptualizations, it comes with limitations, including potential subjectivity in coding and the narrow scope of a single workshop. Our findings should thus be read as a reflective mapping of the field rather than as definitive conclusions.

The remainder of this paper is structured as follows. First, an overview of how behavior and behavioral analytics are defined in various fields is given in Section 2. In the next section, Section 3, the methodology of this paper is discussed. Section 4 discusses the results of this paper, and the paper ends with a discussion and a conclusion in Sections 5 and 6, respectively.

2 Related Work

This section offers a brief overview of how behavior and behavioral analytics are defined or implicitly assumed in the literature of process mining, behavioral

[5] https://www.edba.science

sciences, and information systems. We also examine how prior work across these fields relates behavior to event data, highlighting both explicit definitions and underlying assumptions.

2.1 Behavioral Analysis

Behavior is defined in various ways across research domains, often depending on the entity being studied and the field's focus. Human behavior, for example, has been described as reactions to stimuli [6], sequences of activities [5], or interactions with people [11] or entities [14]. Broader definitions frame behavior as any human activity [23] or simply "the manner of behaving" [8]. Some fields study animal [4], organizational [15], or process behavior [21], while others distinguish between individual and collective behavior [8]. Analytical approaches aimed at understanding, identifying, and anticipating behavioral tendencies include behavior modeling [18], pattern mining [23], and behavioral prediction [21]. In process mining, behavior is typically represented as sequences of activities performed by people or systems [1]. Despite the variation in definitions, common themes include actions, interactions, and responses.

In data-driven behavioral research, three key domains have emerged: Behavior Informatics, Behavior Analytics, and Behavior Computing. Behavior Informatics [5] focuses on modeling, pattern detection, and simulation to uncover hidden mechanisms behind behavior. Behavior Analytics [6] addresses observable behavior through analysis, prediction, and management, while Informatics explains its underlying structure and meaning [9]. Behavior Computing [7] offers a broader framework, embedding behavior in its social, organizational, and environmental context. Serving as an umbrella concept, it integrates insights from Analytics and Informatics while emphasizing contextual influences.

2.2 Relation behavior, events, and event data

Event data captures the occurrence of discrete events, each described by features such as activity, timestamp, and entity. In process mining, such data is structured into event logs that reflect process behavior and are used for tasks like model discovery, behavior prediction, and conformance checking [1]. While process mining primarily focuses on process behavior, it can also take an organizational perspective, analyzing the behavior of resources involved in the process. This includes building social networks, evaluating performance [20], detecting preferences and skills, and examining behavioral patterns over time [17]. When focusing on resources, process mining highlights human behavior as reflected in event data, although it can equally be applied to analyze the behavior of machines, systems, or other entities.

3 Methodology

To gain an initial understanding of how behavior and behavioral analytics are interpreted within the process mining community, we analyze EdbA workshop publications, a forum that reflects emerging discussions in this area, though not necessarily representative of the entire process mining field. We focus on EdbA papers because this workshop is the only venue explicitly centered on behavioral perspectives in process mining, making it a concentrated and relevant corpus for our research question. At the same time, we acknowledge that this choice limits the breadth of perspectives captured, and future work should expand the analysis to other sources of papers to provide a more comprehensive view. Following the principles of the Straussian Grounded Theory Method, we apply a grounded theory approach consisting of open coding, axial coding, and selective coding. This iterative process allows us to inductively identify key concepts, uncover underlying structures, and develop a conceptual understanding grounded in the literature.

3.1 Straussian Grounded Theory

Grounded Theory is a qualitative method aimed at developing theories from empirical data rather than testing existing hypotheses. Among its main variants: Glaserian, Straussian, and Constructivist, we adopt the Straussian Grounded Theory (SGT), which emphasizes integrating existing literature throughout the research process [10,19]. SGT follows a structured, iterative process involving four coding stages: open coding, axial coding, selective coding, and the conditional matrix. This study focuses on the first three, moving from codes to categories to themes to build a theory, in line with Williams and Moser [22]. The fourth stage, which situates findings within broader structural contexts, is beyond the scope of this work.

3.2 Literature Selection

In order to examine how the concepts of behavior and behavioral analytics are defined in the context of the EdbA workshop, we analyze all published papers of this workshop series related to behavior or behavioral analytics. To determine whether the papers from the EdbA workshop are relevant, the following inclusion and exclusion criteria are established. We included papers that were officially published as part of the EdbA workshop at ICPM and use the term "behavior" or "behavioral analytics" in the title, abstract, keywords, or main body. We excluded papers that do not address behavior explicitly, for example, by providing a definition or operationalization of behavior, nor implicitly, such as by focusing on behavioral patterns without labeling them as such. We analyzed all papers presented at the EdbA workshop at ICPM over its full runtime from 2020 to 2024, which was a total of 39 papers. After applying the above-mentioned inclusion and exclusion criteria, 27 papers remained.

3.3 Coding Process

In this section, the following coding process is described. This process aligns with the first three steps of the SGT method [19] and the method described in Williams and Moser [22]. The coding is performed by one coder, the first author.

Open Coding Open coding is the first step in the STG Method and forms the foundation for theory development. In this phase, selected literature is carefully examined and broken down into meaningful data segments [19]. These segments are conceptualized and labeled with inductively derived codes [22]. The goal is to identify patterns and recurring ideas related to behavior and behavioral analytics, creating a structured list of initial codes. In this stage, we discovered 247 different codes from the 27 available papers.

Axial Coding Axial coding is the second phase of the STG Method and builds on the results of open coding. In this step, related codes are grouped into broader categories to reveal underlying structures in the data [19]. While considering factors such as causes, context, actions, and consequences, connections between codes are made more explicit. This process creates a coherent network of concepts [22]. After this stage of coding, the 247 codes found in the previous stage were grouped into 44 code categories.

Selective Coding Selective coding is the final stage of the STG Method, where code categories are refined into main themes [22]. Themes are identified to integrate and connect the results of axial coding, forming a coherent conceptual narrative. Operating at a higher level of abstraction, this phase captures the essence of the studied phenomenon [19], in this case, behavior and behavioral analytics, and sets the stage for theory development. During selective coding, 9 themes were discovered from the 44 code categories.

Example In Figure 1, a short fragment of [3] is shown together with its codings.

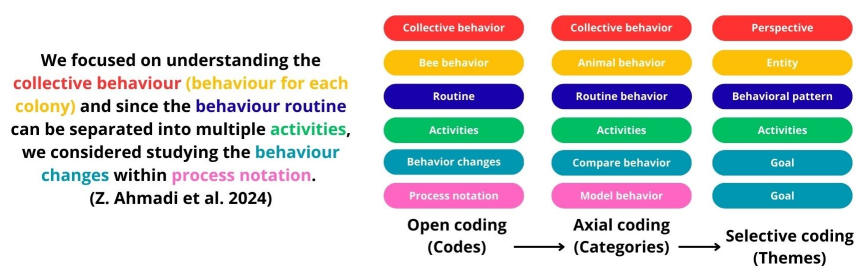

Fig. 1. Example of the coding procedure

In this fragment, six codes were identified during open coding. After axial coding, these were grouped into six categories. Two codes, “Collective behavior” and “Activities”, retained their names. “Bee behavior” was generalized to “Animal behavior”, and “Behavior changes” was grouped under “Compare behavior”, as comparison is needed to identify change. “Process notation” was categorized as “Model behavior”, reflecting its role in modeling behavior. In the final stage, categories were organized into themes. “Collective behavior” was placed under

"Perspective", "Animal behavior" under "Entity", and "Routine behavior" under "Behavioral patterns". "Activities" remained unchanged. Lastly, "Compare behavior" and "Model behavior" were merged into the theme "Goal".

4 Results

From the 39 available papers, we detected that 27 are in some way or form about behavior or behavioral analytics. None of these 27 papers explicitly state what they understand as behavior or behavioral analytics. There are only implicit mentions of what the authors of those papers understand as behavior or behavioral analytics. All implicit mentions of behaviors were coded during the first coding stage, the open coding stage. After open coding the 27 selected papers, 247 different codes emerged. These 247 codes were grouped into 44 code categories during axial coding. In the last coding stage, these 44 code categories were grouped into 9 themes. With these 9 themes, a theory will be constructed around behavior and behavioral analytics. The themes will be highlighted in bold in the text below.

4.1 Behavior

A first theoretical framework that will be constructed from the discovered themes is the Behavior Framework. This framework, showing the most important connections between the relevant themes, is displayed in Figure 2A. The figure and the discovered themes will be explained in the text below.

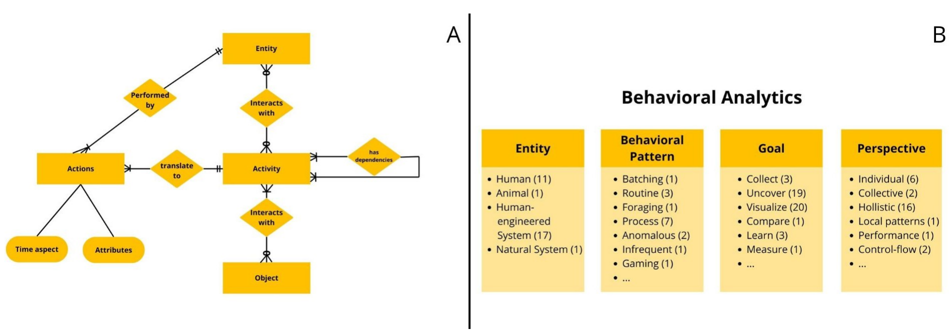

Fig. 2. Behavior and Behavioral Analytics Framework

Behavior within the process mining context can be understood as an activity being performed. An **Activity**, or also addressed as a task or an execution, is a high-level act of behavior. Therefore, this activity consists often of multiple smaller actions. **Actions**, also called events, moves, or state changes, are the smallest unit of behavior. There are atomic instant actions, such as "click a button" or "touch a key on your keyboard". The concept of an atomic action appeared in 15 of the 27 analyzed papers, though often under different names.

Actions can stand alone as activities, but they can also be grouped into higher-level activities. For example, grouping actions together to represent an activity. For example, the actions: "click the number 2 on keyboard" and "click the number 4 on keyboard" are translated to the high-level activity "type age", since the target of both lower-level actions was the same, namely typing the age of the first author of this paper. Each action only belongs to one activity, and an activity has at least one action, but can have more.

We distinguish between actions and activities because behavior in process mining can be studied at different levels of granularity. Actions represent the smallest observable steps, while activities are higher-level interpretations that emerge when several actions are grouped together. For instance, the start and end of an activity are actions, and together they form the activity. Our analysis showed that 14 of the 27 papers described behavior at multiple levels of granularity, distinguishing between fine-grained actions and higher-level activities. In some cases, this hierarchy was made explicit, while in others it was only implied by grouping low-level interactions into broader categories. This indicates that the separation between actions and activities is not our assumption, but rather a recurring way in which behavior is conceptualized in the process mining literature.

Each action has a time aspect that indicates when the action is performed. A **Time Aspect** is presented by a timestamp (day and/or time) such that an order between actions can be established. An action can have multiple timestamps, for instance, a start and end timestamp. The duration of an action is not classified as a time aspect, but is categorized under the other attributes, since we cannot deduce an order of actions by looking at the duration. Besides a time indication, behavior can have other attributes that add extra information about the context in which the behavior occurred. **Attributes** provide context for a behavior, such as location, priority, or purpose. Attributes can be given or derived. For instance, the duration of an action can be derived from a start and end timestamp. Attributes may relate to a specific behavior instance ("high priority" assigned to a support ticket on 8 November) or to a type of behavior (frequency of "Make an offer" across all cases). The former is instance-specific, while the latter summarizes patterns across similar behaviors. The timestamp and the other attributes are inherited from the action to the higher-level activity.

Each action is performed by an entity. The **Entity** can be human, animal, or a system that is either natural (weather) or human-engineered (information system). Each action can only be performed by one entity; however, an entity can perform one or multiple actions. Besides performing actions, entities can also interact with activities. An activity can interact with none, one, or multiple entities, and an entity can interact with none, one, or multiple activities. For instance, the activity "Fill in Loan Application" is performed by a loan officer (human entity) but involves interactions with multiple other entities. During this activity, there are interactions with the client to obtain information to fill in the application (human entity), and there are interactions with an information system in which the information is inputted (human-engineered system).

Besides interactions with entities, there can also be interactions with objects. An **Object** can be interpreted as something on which or with which the behavior is performed. A simple example is eating an apple; the apple is the object of the behavior. In a business context, this can be the document someone is working on. Often, Objects and Entities are seen as the same thing; however, we discuss entities and objects separately to highlight their distinct analytical role. The focus is not on reinterpreting the established term, but on distinguishing between actors that perform behavior (entities) and things that are acted upon (objects). An object can interact with one or more activities, while an activity can interact with none, one, or multiple objects. It's important to note that this is a one-to-many relationship, not a zero-to-many relationship as seen with entities. This distinction is intentional: if an object does not interact with any activity, it does not appear in the dataset. In contrast, entities can exist in the dataset without interacting with an activity, because they may still be recorded as performing the actions that are part of an activity. Lastly, activities have dependencies on one another. These represent control-flow relations such as direct succession, causality, concurrency, or choice. Each activity can have one or multiple dependencies on other activities.

4.2 Behavioral Analytics

Besides our Behavior Framework, we also created a framework describing the act of analyzing behavior, namely the Behavioral Analytics Framework, displayed in 2B. Our Behavioral Analytics Framework consists of 4 dimensions: Entity, Behavioral Pattern, Goal, and Perspective. Any Behavioral analytics endeavor is built from these 4 dimensions. The numbers beside the examples of these dimensions represent their occurrence. Note: Some papers included multiple use cases involving different focuses per dimension. The four dimensions of the Behavioral Analytics Framework were derived inductively from the literature and refined to capture the essential aspects of how behavior is analyzed. They emerged consistently across the literature, but they also reflect conceptual reasoning: to analyze behavior, one must specify the actor (Entity), the phenomenon of interest (Behavioral Pattern), the purpose of analysis (Goal), and the viewpoint (Perspective). This reasoning guided our choice to structure the framework around these four dimensions.

The first dimension is the **Entity** dimension. This dimension represents the entity performing the behavior. There are four types of entities, just as seen in the Behavior framework: human, animal, human-engineered system, and natural system. Depending on the data, this can be one type of entity or a combination. For instance, human-computer interaction data is a type of behavioral data that has two types of entities: a human and a human-engineered system. In the analyzed literature set, one paper focused on animals as the entity, and another on a natural system, specifically the weather. Additionally, 13 papers involved human entities, while 17 papers featured human-engineered systems as entities.

Behavioral Patterns form the second dimension of the framework. This dimension captures the phenomenon of interest, which is necessary to frame the scope of the analysis. Different patterns require different methods. Common examples in the literature include routine behavior, anomalous behavior, daily activities, and batching behavior. Across the workshop papers, 18 distinct behavioral patterns were identified, including foraging behavior, switch behavior, collaborative work, and user journeys. Among these, four patterns appeared in multiple papers: process behavior (7), routine behavior (3), anomalous behavior (2), and daily behavior (2).

The third framework dimension is the **Goal** of the behavioral analytics endeavor: the intended outcome of analyzing behavior. Goals vary and often overlap, including collecting behavioral data, uncovering and visualizing behavior, comparing behavior, learning patterns, and measuring behavior. A common combination is uncovering and visualizing behavior, first identifying patterns such as routines or infrequent behavior, then presenting them in interpretable formats like process models or sequence diagrams. Collecting behavior refers to gathering data through observation, instrumentation, or simulation, while learning patterns emphasizes discovering structures or trends. Measuring behavior involves quantifying aspects such as frequency, duration, fitness, or complexity. This dimension highlights that analyses are purpose-driven: the same data may serve different goals, and making these goals explicit clarifies method choices.

Among the analyzed papers, the two most common goals were visualizing behavior (20) and uncovering behavior (19). Less frequently, collecting and learning behavior each appeared in 3 papers, while comparing and measuring behavior were each found in 1 paper.

The final dimension of the framework is the **Perspective** taken during the analysis: from which angle or viewpoint is the behavior examined? Perspective is included because the same behavior can be understood differently depending on the chosen lens. This dimension reflects the analyst's standpoint and determines which aspects of behavior are emphasized. Common perspectives include control-flow, performance, organizational, individual, collective, holistic, and local patterns. A variety of perspectives were explored in the workshop papers. The most frequently adopted was the holistic perspective (16), followed by the individual perspective (6). The collective and control-flow perspectives were each observed in 2 papers, while the performance and local perspectives appeared once each.

5 Discussion

5.1 Reflections on Behavioral Analytics in Process Mining

Behavioral analytics in process mining is a diverse and emerging field that spans a wide range of entity types, behavioral patterns, goals, and analytical perspectives. While the majority of studies focus on human or human-engineered system behavior, there is evidence of interest in non-human entities such as animals and natural systems. A rich set of behavioral patterns has been explored, ranging from foraging and collaboration to user journeys, with recurring emphasis on process, routine, anomalous, and daily behaviors. The primary aims across studies are to visualize and uncover behavior, reflecting a strong interest in descriptive and exploratory analysis. Furthermore, most studies adopt a holistic perspective, indicating a tendency to analyze behavior at the level of systems or populations rather than individuals or process fragments. Together, these trends suggest that behavioral analytics in process mining is not only broad in scope but also still evolving, with opportunities for further refinement in theoretical framing, methodological approaches, and application contexts.

5.2 Implications

This study aims to contribute to a clearer and more structured understanding of how the concepts of behavior and behavioral analytics are interpreted within the process mining community. Through a qualitative analysis of EdbA publications, we identified 9 common themes and created two frameworks highlighting the relations between these different themes. These frameworks represent an intermediary step toward a shared vocabulary and open the door for more coherent theory development in future work, rather than providing a definitive account of behavior in process mining.

At the same time, this study faces several limitations. It is restricted to a single venue (EdbA), offering a focused but incomplete view of the community's perspectives. The qualitative coding process also involves interpretive subjectivity, so alternative categorizations are possible. The proposed themes and frameworks should therefore be seen as exploratory rather than consolidated, highlighting current tendencies rather than definitive definitions. Despite these threats to validity, the study illustrates the potential value of a clearer conceptualization of behavior in process mining. A shared vocabulary can reduce confusion, enable more robust comparisons, and guide the design of analysis techniques tailored to aspects such as control-flow, decision-making, user interactions, or organizational dynamics.

Practically, the insights from this study can support the design of behavior-aware tools by clarifying which behavioral perspectives analysts prioritize, such as identifying unusual paths or spotting bottlenecks over time, which can inform interface design or automated pattern suggestions. Additionally, these insights can aid in the training of novice analysts by offering a clearer vocabulary and conceptual structure to describe and interpret behavioral patterns in event data, for example, through targeted learning modules or annotated process examples. Finally, at the community level, this work encourages a more reflective and interdisciplinary discourse around the use of behavioral concepts in process mining. It invites researchers to make their assumptions explicit and consider how definitions of behavior influence both research outcomes and practical applications.

6 Conclusion

This study provides a first systematic exploration of how the concepts of behavior and behavioral analytics are used and understood within the process mining community, with a particular focus on contributions from the EdbA workshop. Through open, axial, and selective coding, we identified 9 major themes that are addressed and how they are interconnected with one another.

Behavior within the process mining context can be understood as an activity being performed. An activity consists of one or multiple actions with a time aspect and other attributes describing the context. This activity is performed by an entity and can have interactions with other entities or objects. The activity also has dependencies on other activities. Behavioral Analytics is the analysis of behavior along four key dimensions: entity, behavioral pattern, goal, and perspective. Each analysis combines elements from these dimensions. The entity is who or what performs the behavior. The behavioral pattern refers to the type of behavior studied, such as routine, gaming, or foraging. The goal defines what the analysis aims to achieve, like collecting, visualizing, or comparing behavior. Finally, the perspective reflects the angle from which the behavior is examined.

The contribution of this work lies in providing an intermediary step toward a shared vocabulary and stronger theoretical and methodological alignment in behavior-focused process mining research. At the same time, several limitations must be acknowledged. The study is restricted to papers from a single workshop, and the qualitative coding process inherently involves interpretive subjectivity. Consequently, the findings should be interpreted as exploratory rather than definitive. Future studies can build on this foundation by broadening the scope to include a wider range of literature, such as main track ICPM papers, journals, and industry reports, to explore how conceptualizations of behavior vary across contexts and over time. Moreover, complementary research methods, such as interviews, surveys, or Delphi studies with researchers and practitioners, can help validate, refine, and expand the themes identified here.

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